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Original Research

Application of the wireless wearable monitoring system improves the turnaround time in gastrointestinal endoscopy center

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Abstract

Objectives: To investigate whether the wireless monitoring system can reduce the turnaround time in gastrointestinal endoscopy center.

Methods: Enrolled patients were randomly allocated to the study or control group. The primary outcome was the turnaround time. The secondary outcomes included the total time from entering examination room to discharge (T1), the total time of monitoring mean arterial pressure (T5), incidences of hypotension, bradycardia, and hypoxia, evaluation of the medical staff, and rate of effective monitoring.

Results: 166 patients were analyzed. The turnaround time of study group was significantly shorter than that of control group (1.49 ± 1.82 vs. 4.78 ± 3.77 min, $P < .001$). The study group had a significantly shorter T1 (44.69 ± 4.54 vs. 49.49 ± 5.53 min, $P < .001$), but a longer T5 (44.39 ± 4.38 vs. 42.21 ± 4.97 min, $P = .012$). The medical staff gave higher scores to the wireless monitoring system in terms of practicability and application value ($P < .05$), but gave equally high evaluations on data reliability ($P > .05$). The rate of effective monitoring was acceptable in both groups.

Conclusion: Application of the wireless monitoring system in an endoscopy center is feasible and acceptable, and improves the turnaround time.

Trial registration: This study was registered at the Chinese Clinical Trial Registry on 19 April 2024 (ChiCTR2400083277).

Key messages:

What is already known on this topic?

- Wireless wearable monitoring system has been applied to the wards.

What this study adds?

- Wireless wearable monitoring system can be used in endoscopy centers and can reduce the turnaround time.

How this study might affect research, practice, or policy?

- This system may further improve the safety of anesthesia by global monitoring.

Keywords: wireless wearable monitoring system; wired monitor; painless gastrointestinal endoscopy; turnaround time; application value

Introduction

Gastrointestinal endoscopy is a common but effective operation to diagnose and treat diseases of the digestive tract. Advancements in this technology and increased public health awareness have led to a continuous rise in gastrointestinal endoscopy cases. In some areas, the increase rate is over 10% annually [1]. A national survey conducted in 3714 hospitals in China showed that 3562.50 (1299.75, 8426.75) cases of gastrointestinal endoscopy were performed at these hospitals in 2020, while the number of endoscopists was only 4 (2, 7) [2]. Moreover, the shortage of anesthesiologists is widespread in countries at different levels of development [3, 4]. The mismatch between large number of endoscopy cases and shortage of doctors leads to a mammoth workload for the endoscopists, anesthesiologists, and nurse anesthetists. Thus, it is important to improve the turnaround time and ensure safety at the same time.

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In the traditional endoscopy center, wired monitors are commonly used to monitor vital signs. Monitoring begins after entering the examination room and resumes in the recovery room. Repeated connection of wired monitors increases the workload of medical staff and may decrease work efficiency. In addition, the surveillance monitoring gap during the transit process is very dangerous for high-risk patients.

Wireless wearable monitors integrate wireless communication technology, sensors, flexible electronics, and other technologies. They have the function of detecting or monitoring body signs and their changes. The wireless feature expands the application scenario from medical institutions to society. Basically, continuous monitoring of vital signs with wireless devices in wards is believed to improve patient care and alarm severe adverse events, and reduce the workload of medical personnel [5, 6].

Continuous monitoring with wireless monitors has been investigated in general wards, surgical wards, and emergency wards [5, 7, 8]. However, there are rare studies focusing on its application in gastrointestinal endoscopy center. In the present study, we plan to establish a vital sign monitoring system based on wireless devices, and investigate whether it can shorten the turnaround time and enhance the satisfaction of medical workers.

Patients and methods

Overview of this study

This is a comparative effectiveness study. The study was approved by the ethics committee and performed in Chengdu Second People's Hospital. The ethics approval number is 2023603. Before enrolling subjects, the research team got a register number (ChiCTR2400083277) at the Chinese Clinical Trial Registry. All subjects signed an informed consent before grouping. The investigators declare that they performed this study in line with the principles of the Declaration of Helsinki.

Patients

Patients undergoing elective painless gastroenteroscopy in Chengdu Second People's Hospital were screened. The inclusion criteria included: (i) patients who are ≥18 years old; (ii) patients who understand the study and sign the informed consent voluntarily. The exclusion criteria were as follows: (i) severe hypertension; (ii) severe cardiopulmonary disease rendering patients unable to tolerate endoscopy; (iii) suspected shock, gastrointestinal perforation, or other critical conditions; (iv) patients with uncontrolled mental illness, and those who cannot cooperate with the study procedure.

Construction of the wireless wearable monitor system

The wireless wearable monitoring device (CareClever® Wireless Multi-Parameter Patient Monitoring System, MC10 Series, Raiing Medical Company, China) consists of three parts, namely vital sign sensors, signal repeater, and monitoring terminal. The vital sign sensors include an electrocardiogram (ECG) collector, pulse oxygen saturation (SpO_2) module, and noninvasive blood pressure (NBP) module, which can continuously monitor heart rate (HR) and blood oxygen saturation in real time (collecting data once per second), and automatically measure blood pressure at a certain time. The signal repeater receives the vital signs through Bluetooth and then transmits the data to the monitoring terminal through Wireless Fidelity. The monitoring terminal includes multi-screen interactive system terminals and handheld monitoring terminals (personal digital assistant, PDA), which can be used to view monitoring data and receive alarm prompts. This device is shown in Fig. 1.

The core of this technology is the deployment of the Internet of Things (IoT) covering the whole area in the digestive endoscopy center, and the corresponding sensors, portable terminals, and display terminals for wearable patient monitoring. The core of integration system of process management is the integration of devices and clinical procedures. The technology route of constructing a wearable, global, and full-time monitoring system is depicted in Fig. 2.

Application process of the wireless wearable monitor

After patients enter the waiting area, nurse anesthetists enter basic information of the patients with PDA, attach the modules of ECG, SpO_2, and NBP to patients, and connected signal repeater and sensors using Bluetooth. Upon completing the above steps, the patients' basic vital signs can be shown on the PDA. In addition, the PDA can receive technical alarms (e.g. poor equipment wearing quality or equipment failure) and physiological alarms (physiological parameters exceeding the normal range). The devices were removed at the end of study.

Randomization and grouping

Enrolled patients were randomly allocated to the study group or control group according to the random number table. In the study group, nurse anesthetists put the wireless monitoring devices on patients in the waiting area and took them down when patients met the discharge criteria. In the control group, anesthesiologists put the traditional monitor on patients in the examination room and took them down at the end of examination. Then, nurse anesthetists put the traditional monitor on patients in the post-anesthesia care unit (PACU) and took them down when patients met the discharge criteria. The anesthesiologists for anesthesia and the nurse anesthetists in PACU were blinded to the objective of this study.

Study procedure

After trial registration, recruitment posters were pasted on the walls of the endoscopy center and the anesthesia assessment office. Investigators thoroughly introduced the study to eligible patients during pre-anesthesia assessment. Upon signing the informed consent, patients were enrolled, and they could withdraw anytime.

On the examination day, investigators randomly allocated the patients according to the order of registration. Monitoring of vital signs was performed based on the grouping information. Patients in both groups received the same anesthesia plan. In the waiting area, butorphanol 10–20 μg/kg was administered intravenously in advance. Pure oxygen was supplied continuously at a rate of 5 l/min via a nasal cannula in the examination room. Anesthesia was initiated with propofol 2 mg/kg when SpO_2 reached 100%. When the bispectral index (BIS) value was below 60, gastrointestinal endoscopy was started. Top-ups were administered to control the BIS value between 40 and 60. Dopamine 2 mg was injected to improve hypotension (defined as mean arterial pressure, MAP <65 mmHg or > 20% reduction from baseline). Atropine 0.5 mg was injected when the HR was below 45 beats/min. Manual assisted ventilation was applied to improve hypoxia ($SpO_2 < 90\%$). When the endoscopy finished, patients were transferred to the PACU for recovery.

At the end of each day, the endoscopists, anesthesiologists, and nurse anesthetists marked the two monitoring methods in terms

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Figure 1 Components of the wireless monitoring system. a. Wireless blood pressure module; b. wireless wrist pulse oximeter; c. wireless ECG sensor; d. wireless body temperature sensor; e. vital signs relay unit (patient-end hub); f. handheld monitoring terminal; g. bedside monitoring terminal; h. mobile-vehicle monitoring terminal; i. central monitoring terminal

of the practicability, reliability, and application value. The scores ranged from 0 to 10 (higher scores indicating better performance).

During the study process, vital signs were documented every 3 min. The time of entering examination room, initiating anesthesia, initiating operation, finishing operation, getting out of examination room, entering PACU, and discharge were accurately recorded.

Definitions

(i) T1: The total time from entering examination room to discharge;
(ii) T2: Operation time = the time of finishing operation discharges. The time of entering examination.
(iii) T3: Recovery time = the time of discharge – the time of entering PACU;
(iv) T4: Estimated turnaround time = T1scharge.
(v) T5: The total time of monitoring MAP = the time of last MAP – the time of first MAP (study group) or = (the time of last MAP – the time of first MAP in PACU) + (the time of last MAP – the time of entering examination).
(vi) Rate of effective monitoring = (the number of vital sign values/the number of measurements) x 100%. The data were extracted from the central server.

Outcomes

The primary outcome was the estimated turnaround time (T4). The secondary outcomes included: (i) The total time from entering examination room to discharge (T1), (ii) the total time of monitoring MAP (T5), (iii) incidences of hypotension, bradycardia, and hypoxia, (iv) evaluation of the endoscopists, anesthesiologists and nurse anesthetists, and (v) rate of effective monitoring.

Statistical analysis

Calculation of sample size was based on the time for wearing the monitors. In our preliminary observation, time for wearing a wireless wearable monitor was 90 seconds on average, while wearing a traditional monitor in operation room and PACU took 200 seconds on average. With a significance level (α) of 0.05, power (1-β) of 90%, and an estimated 5% dropout rate, a minimum of 80 participants per group was required in each group.

Data were entered and analyzed by two independent investigators using SPSS 26.0. First, the normality of the continuous variables was validated by the Kolmogorov–Smirnov test. Normal data were shown as mean ± standard deviation (SD), and statistical analyses were performed using *t*-test. Non-normal data were presented as median (25th quantile, 75th quantile) and statistical analyses were performed using Wilcoxon rank-sum test. Count data, for instance, the incidence of adverse events, were shown as numbers (percentages) and analyzed using chi-square test. $P < .05$ was considered statistically significant in this study.

Results

From 22 April to 1 November 2024, 182 patients signed the informed consent, 16 of them withdrew before examination. Thus, 166 patients (45.16 ± 9.58 years, 44.58% male) were randomly allocated into the 2 groups. No patient dropped out after grouping. Flowchart of this study is shown in Fig. 3. Demographic data, including sex, age, and body mass index (BMI), were comparable between the groups (Table 1). Baseline data, including the comorbidities, drug allergy history, endoscopy type, HR, MAP, and SpO_2, had no significant difference between the groups (Table 1).

In the study group, the total time from entering examination room to discharge was significantly shorter in comparison with the control group (44.69 ± 4.54 vs. 49.49 ± 5.53 min, $P < .001$) (Table 2). However, the operation time and recovery time were comparable between the groups (15.45 ± 3.06 vs. 16.36 ± 4.29 min, $P = .116$; 27.75 ± 3.27 vs. 28.35 ± 2.87 min, $P = .105$, respectively) (Table 2). Therefore, the estimated turnaround time was significantly shorter in the study group than in the control group

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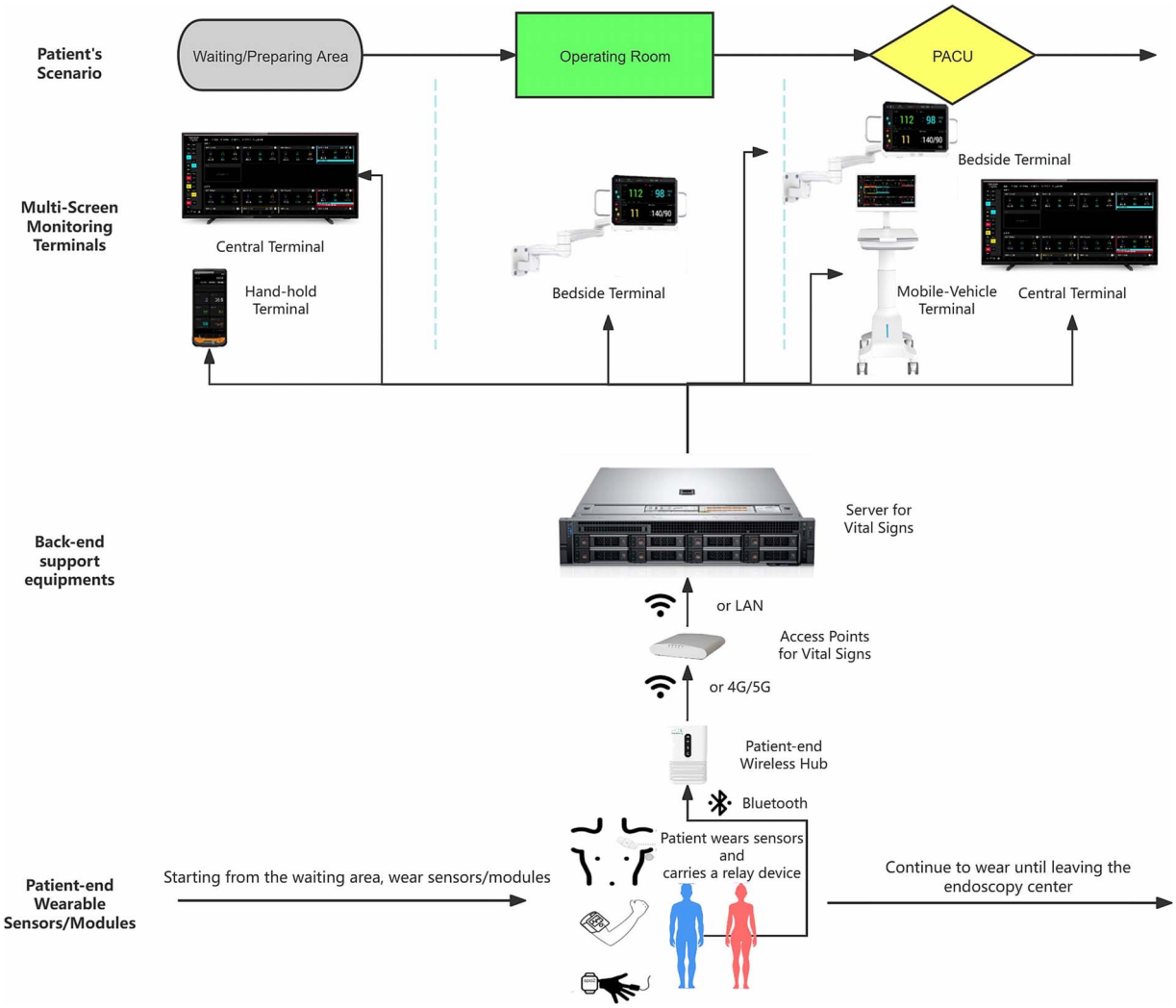

Figure 2 The technology route of constructing wearable, global, and full-time monitoring system

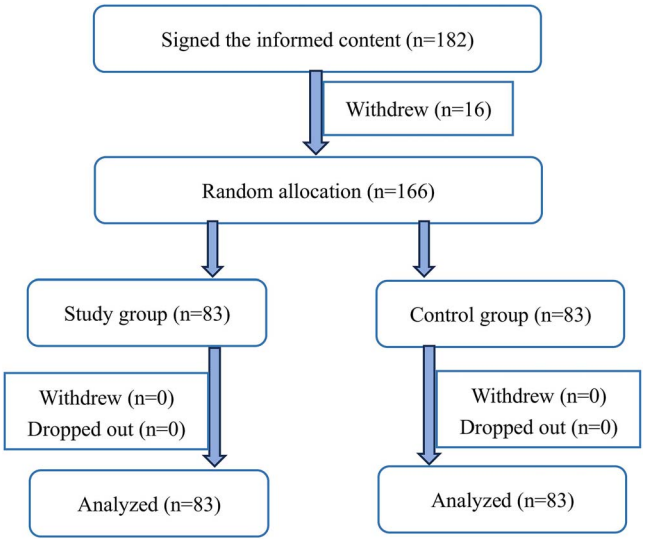

Figure 3 Flowchart of this study

(1.49 ± 1.82 vs. 4.78 ± 3.77 min, $P < .001$) (Table 2). By reviewing records in the monitoring devices, we calculated the total time of monitoring MAP. The results showed that monitoring duration was significantly longer in the study group (44.39 ± 4.38 vs. 42.21 ± 4.97 min, $P = .012$) (Table 2).

As for the safety of anesthesia, the doses of propofol and butorphanol were of no significance between the groups ($P = .098$, $P = .512$, respectively) (Table 2). The cases of hypotension, bradycardia, and hypoxia reflecting by the case report forms, were similar in the groups ($P = .542$) (Table 2).

Table 2 shows the satisfaction scores of the endoscopists, anesthesiologists, and nurse anesthetists. All of them thought that practicability and application value of the wireless monitoring device were higher than that of the traditional monitor ($P < .05$), while the reliability was equally high for both devices ($P > .05$).

The rates of effective monitoring of HR, blood pressure (BP), and SpO_2 are shown in Table 3. The rates of HR and BP among preparation area, operation room, and PACU were significantly different ($P < .001$, $P = .002$, respectively), while the rates of SpO_2 were similar in the three areas ($P = .087$).

Discussion

In the present study, we conducted a randomized parallel-designed study to investigate whether the wireless monitoring device can reduce the turnaround time in gastrointestinal center.

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Table 1. Demographic data and baseline of the patients.

Parameters	Study group	Control group	P
Sex (*n*, %)			
Male	38, 45.78	36, 43.37	.876
Female	45, 54.22	47, 46.63	
Age, years	47.19 ± 13.72	44.45 ± 13.30	.150
BMI, kg/m^2	23.67 ± 4.46	22.90 ± 3.43	.165
Comorbidity (*n*, %)			
Yes	15, 18.07	14, 16.87	1.000
No	68, 81.93	69, 83.13	
Drug allergy history (*n*, %)			
Yes	3, 3.61	5, 6.02	.720
No	80, 96.39	78, 83.98	
MAP, mmHg	76.34 ± 9.37	77.58 ± 10.15	.102
HR, beats/min	72.34 ± 10.12	69.28 ± 9.74	.094
SpO_2, %	98.78 ± 5.24	98.26 ± 6.13	.453
Endoscopy type (*n*, %)			
Gastric endoscopy	3, 3.61	4, 4,82	.754
Intestinal endoscopy	13, 15.66	10, 12.05	
Gastrointestinal endoscopy	67, 80.72	69, 83.13	

BMI, body mass index. MAP, mean arterial pressure. HR, heart rate. SpO_2, pulse oxygen saturation.

The results indicated the advantages of this kind of device over a traditional monitor.

For gastrointestinal endoscopy under sedation or anesthesia, vital signs including blood pressure, ECG, and SpO_2 should be monitored [9]. Traditional monitors are wired. It needs to be taken on and off in both examination room and PACU, which results in extra time and impacts the work efficiency. Our results show that the estimated turnaround time is significantly shortened by applying the wireless device. Since it takes significant manpower to do the timekeeping of taking the monitors on or off in clinical practice, we estimated the turnaround time by the total time from entering examination room to discharge minus operation time and recovery time. However, it is difficult to determine whether this device can improve the work efficiency in gastrointestinal center. There are two measures for describing work efficiency, namely the number of cases completed in the same time and the time to perform the same number of cases. Both metrics are challenging to precisely calculate, since they are mainly influenced by the number of patients, the types of endoscopies, whether the procedure was technically challenging, and the efficiency of endoscope disinfection. Nevertheless, we interviewed the endoscopists, anesthesiologists, and nurse anesthetists in terms of the practicability, data reliability, and application value of the wireless device according to their practice. All of them gave high evaluation to it, which indirectly shows the facilitation.

The application of wireless monitors has been well investigated in the wards, especially in surgical departments. Downey *et al.* [10] observed the practicability of wireless monitors in 226 surgical patients and most of them thought it acceptable as for comfort and felt an increased sense of security. After surgery, patients are encouraged to get out of bed early under the principles of enhanced recovery after surgery (ERAS) [11]. However, postoperative monitoring of vital signs has been a challenge in clinical practice. Although traditional wired devices can play a necessary role in monitoring, in practice, they may also pose a certain limitation to the patient's range of motion. Mei *et al.* [12] demonstrated high accuracy and reliability of wireless wearable monitoring devices in postoperative monitoring when compared with traditional wired devices. In addition, some studies have reported the potential of wireless wearable monitoring devices to alarm deterioration of patients [13, 14]. In this study, detection rate of adverse vital signs is comparable between the groups. This can be attributed to the low ASA grades of patients and the safe anesthesia medication in our study. However, post-anesthesia transport is associated with adverse events, especially for high-risk patients [15]. Future studies can investigate the importance of wireless monitors in detecting abnormal vital signs and reducing adverse events of high-risk patients in an endoscopy center.

According to the European guideline, trained non-anesthesiologists are permitted to provide sedation for gastrointestinal endoscopy [16, 17]. Nurse anesthetists have widely participated in the perioperative management under the guidance of anesthesiologists, which alleviates the problem of insufficient anesthesiologists. By applying the wireless monitor system, one attending anesthesiologist can simultaneously monitor several patients. This practice improves the safety of sedation provided by nurse anesthetists and junior anesthesiologists. In the present study, the evaluations of the medical staff of the wireless system partly evidenced this practice. Further studies can be conducted to confirm this issue.

In this study, the rates of effective monitoring of HR and BP among different situations have a considerable discrepancy, but the rates of SpO_2 show a high consistency. The measurement modes of vital signs should be responsible for these results. The lateral position may reduce the contact between the ECG electrode and the skin, and further lead to relatively larger data loss during operation. Nevertheless, the ECG sensor collects data once a second, which greatly counteracts the influence of data loss. Movements of the upper arm can break in BP measurement. Thus, the rate of effective monitoring in PACU would be declined by recovering patients. In addition, both traditional monitor and the wireless system have to face this problem. Unfortunately, we cannot obtain the rates of traditional monitors, since it is unrealistic to input patients' information into the monitors in busy clinical settings. So, we do not compare the difference between traditional monitor and the wireless system in terms of the rate of effective monitoring.

As for the economic feasibility, we have made a cost analysis. The total cost of the wearable system included initial purchase investment (about $249 600 for a work station with 60 sets of vital-signs monitors), maintenance fee (about $2300 per year), and the costs of consumables (such as blood oxygen probes and blood pressure cuffs) (about $1800 per year). Over a 10-year lifespan of the system, the annualized cost was $36 430, assuming a 5% discount rate. The annualized cost per monitoring unit amounts to $607 after allocating shared expenditures for central workstations, servers, and network infrastructure. Compared to conventional desktop monitors procured by an equivalent healthcare institution, the average cost was approximately $945. Therefore, the wearable monitoring system demonstrates a 35.8% reduction in annualized cost theoretically.

There are some limitations in this study. First, we failed to demonstrate whether the reduced turnaround time can increase the work efficiency in endoscopy center. As previously discussed, many factors can influence work efficiency. Thus, it is difficult to measure it in the real world. Second, we used an indirect method to calculate the turnaround time, which must be influenced by the accuracy of documented time points. To reduce the impacts as much as possible, the recorders were required to record the time to a precision of seconds. In addition, an investigator of this study randomly checked the reliability of case report forms and

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Table 2. Comparison of the main outcomes between the study group and control group.

Parameters	Study group	Control group	P
The time			
T1, min		44.69 ± 4.54	49.49 ± 5.53	<.001
T2, min		15.45 ± 3.06	16.36 ± 4.29	.116
T3, min		27.75 ± 3.27	28.35 ± 2.87	.105
T4, min		1.49 ± 1.82	4.78 ± 3.77	<.001
T5, min		44.39 n3.777	42.21 n3.777	.012
Safety of anesthesia				
Butorphanol, mg	0.32 phanol	0.33 phanol	.512	
Propofol, mg	244.44 l, mges	246.28 l, mges	.098	
Adverse events, *n*				
Hypotension	20	20	.542	
Bradycardia	10	5		
Hypoxia	7	6		
Satisfaction scores of the endoscopists, anesthesiologists and nurse anesthetists				
Endoscopists				
Practicability	9 (9, 10)	8 (7, 9)	.002	
Reliability	9 (9, 10)	9 (9, 10)	.543	
Application value	10 (9, 10)	7 (6, 8)	<.001	
Anesthesiologists				
Practicability	10 (9, 10)	8 (8, 9)	<.001	
Reliability	9 (9, 10)	9 (9, 10)	.457	
Application value	10 (10, 10)	8 (8, 8)	<.001	
Nurse anesthetists				
Practicability	10 (9, 10)	8 (8, 8)	<.001	
Reliability	9 (9, 10)	9 (9, 10)	.513	
Application value	10 (10, 10)	8 (8, 8)	<.001	

T1, total time, = the time of discharge - the time of entering examination room. T2, operation time. T3, recovery time, = the time of discharge - the time of entering PACU. T4, estimated turnaround time, = T1 - T2 -T3. T5, total time of monitoring mean arterial pressure.

Table 3. Rate of effective monitoring of the wireless monitoring system.

Parameters	Preparation room	Operation room	PACU	*p*
Heart rate	98.25 (98.12, 98.89) %	92.48 (90.32, 96.88) %	97.78 (96.69, 98.20) %	<.001
Blood pressure	91.67 (90.00, 100) %	100 (90.00, 100) %	88.88 (80.00, 100) %	.002
SpO_2	99.78 (98.89, 99.88) %	99.43 (99.10, 99.92) %	98.66 (98.12, 99.10) %	.087

confirmed the accuracy. Lastly, we did not collect the evaluation of patients.

Conclusions

Application of the wireless wearable monitoring system in an endoscopy center is feasible and acceptable. The wireless system is associated with shorter turnaround time and higher evaluation of the medical staff. Further studies can attempt to determine the effects of the wireless system on work efficiency, high-risk patients, and human resource allocation measures to alleviate the anesthesiologist shortage.

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Author contributions

Bo Zhu, Yu Huang, Min Wang, and Ling Liu were fully or partly involved in the conception and design, or analysis and interpretation of the data; the drafting of the paper or revising it critically for intellectual content. All authors have approved the final approval of version to be published and agreed to be accountable for all aspects of the work.

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Data availability

The data underlying this article are available in the article.

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